

SAADUDDIN MAHMUD

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PhD Candidate (ABD) — NIW Completed

PERSONAL STATEMENT

Computer Science Ph.D. candidate with 5+ years of expertise in Agentic AI. My research focuses on aligning AI agents with human feedback using scalable policy optimization and reward learning methods. I also bring extensive experience designing and analyzing cooperative and strategic multi-party frameworks.

EDUCATION

University of Massachusetts Amherst <i>Ph.D. in Computer Science</i>	MA, USA <i>September 2021 – November 2026 (Expected)</i>
University of Massachusetts Amherst <i>M.S. in Computer Science (CGPA: 4.00/4.00)</i>	MA, USA <i>September 2021 – May 2024</i>
University of Dhaka <i>B.S. in Computer Science and Engineering</i>	Dhaka, Bangladesh <i>January 2016 – December 2019</i>

EXPERIENCE

AI/ML Intern <i>Adobe Incorporated</i> Incoming intern with the Adobe Firefly team, working on the Firefly family of models.	May 2026 – August 2026 <i>Seattle, WA</i>
AI Research Intern <i>Nissan Group North America, Incorporated</i> Engineered a personalized autonomous routing framework by leveraging LLMs to convert individual natural language requests into custom reward functions, validated via real-world deployment on a prototype vehicle.	June 2024 – August 2024 <i>San Jose, CA</i>
AI Research Intern <i>Nissan Group North America, Incorporated</i> Built interpretability and explainability methods for autonomous driving agents. Worked with the ROS stack of a prototype vehicle, analyzing driving data and Level 3 decision systems to generate and visualize explanations for driving decisions at 10Hz speed.	June 2023 – September 2023 <i>San Jose, CA</i>

SKILLS

Research: Reinforcement Learning, Planning, Alignment, Multi-Agent Systems, Game Theory, Large Language Models, Vision-Language-Action Models.

Programming: Python, C++, C, Julia, CUDA.

PATENTS

- Dynamic Refinement of Custom Classes Using Zero-Shot Images. *(Under review)*.
- Vehicle Decision Making Using Sequential Information Probing. *US Patent App. 18/429,196*.

SELECTED PEER-REVIEWED PUBLICATIONS

- Inference-Aware Prompt Optimization for Aligning Black-Box Large Language Models.
Saaduddin Mahmud, Mason Nakamura, Kyle Hollins Wray, Shlomo Zilberstein. (**AAAI, 2026**).
- MAPLE: A Framework for Active Preference Learning Guided by Large Language Models.
Saaduddin Mahmud, Mason Nakamura, and Shlomo Zilberstein. (**AAAI, 2025**).
- Causal Explanations for Sequential Decision Making Under Uncertainty: Foundations and Analysis.
Samer B. Nashed, **Saaduddin Mahmud**, Claudia V. Goldman, and Shlomo Zilberstein. (**Journal of Artificial Intelligence Research, 2025**).

4. Explaining the Behavior of POMDP-based Agents Through the Impact of Counterfactual Information.
Saaduddin Mahmud, Marcell VazquezChanlatte, Stefan Witwicki and Shlomo Zilberstein. (**AAMAS, 2024**).
5. Learning Constraints on Autonomous Behavior from Proactive Feedback.
Connor Basich*, **Saaduddin Mahmud***, and Shlomo Zilberstein. (**IROS, 2023**).
6. Explanation-Guided Reward Alignment.
Saaduddin Mahmud, Sandhya Saisubramanian, and Shlomo Zilberstein. (**IJCAI, 2023**).
7. Semi-Autonomous Systems with Contextual Competence Awareness.
Saaduddin Mahmud, Connor Basich, and Shlomo Zilberstein. (**AAMAS, 2023**).
8. Learning Optimal Temperature Region for Solving Mixed Integer Functional DCOPs.
Saaduddin Mahmud, Md. Mosaddek Khan, Moumita Choudhury, Long Tran-Thanh, Nicholas R. Jennings. (**IJCAI, 2021**).
9. AED: An Anytime Evolutionary DCOP Algorithm.
Saaduddin Mahmud, Moumita Choudhury, Md. Mosaddek Khan, Long Tran-Thanh, Nicholas R. Jennings. (**AAMAS, 2020**).
10. A Particle Swarm Based Algorithm for Functional Distributed Constraint Optimization Problems.
Moumita Choudhury, **Saaduddin Mahmud**, Md. Mosaddek Khan. (**AAAI, 2020**).

PREPRINTS, WORKSHOP & UNDER REVIEW

1. CoLLAB: A Framework for Designing Scalable Benchmarks for Agentic LLMs.
Saaduddin Mahmud, Eugene Bagdasarian, Shlomo Zilberstein. (**Scaling Environments for Agents Workshop at NeurIPS, 2025**).
2. Colosseum: Auditing Collusion in Cooperative Multi-Agent Systems.
Mason Nakamura, Abhinav Kumar, Saswat Das, Sahar Abdelnabi, **Saaduddin Mahmud**, Ferdinando Fioretto, Shlomo Zilberstein, Eugene Bagdasarian. (Under review).
3. Verification Required: The Impact of Information Credibility on AI Persuasion.
Saaduddin Mahmud, Eugene Bagdasarian, Shlomo Zilberstein. (Under review).
4. Terrarium: Revisiting the Blackboard for Studying Multi-agent Attacks.
Mason Nakamura, Abhinav Kumar, **Saaduddin Mahmud**, Sahar Abdelnabi, Shlomo Zilberstein, Eugene Bagdasarian. (Under review).

HONORS, AWARDS & PROFESSIONAL SERVICE

Distinguished PhD Candidate	2024
<i>Recognition for outstanding performance in Ph.D. candidacy qualification.</i>	
Conference Reviewing	2024-Current
<i>AAAI, ICML, and IJCAI.</i>	
B.Sc. Scholarship from the University Grants Commission	2021
<i>Awarded for outstanding performance in B.Sc.</i>	